

A Counterexample to the κ -Octahedral $\ell_1(\kappa^+)$ Question

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Status. Candidate counterexample, likely valid subject to human review. The packet answers Question 6.2 of Ciaci–Langemets–Lissitsin [1] negatively.

1 Source Question

The source is Stefano Ciaci, Johann Langemets, and Aleksei Lissitsin, *A Characterization of Banach Spaces Containing $\ell_1(\kappa)$ via ball-covering properties*, arXiv:2102.13525 [1]. Question 6.2 appears on page 15 of the compiled source paper:

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since the failure of the α -BCP $_{\kappa}$ for all $\alpha \in (-1, 1)$ implies κ -octahedrality, but Theorem 5.13 suggests that the converse might not hold.

Question 6.2. *Let κ be an infinite cardinal. If a Banach space is κ -octahedral, then does it contain an isomorphic copy of $\ell_1(\kappa^+)$?*

In particular, we do not know the answers to these questions even in the case $\kappa = \omega$.

The question asks whether, for an infinite cardinal κ , every κ -octahedral Banach space contains an isomorphic copy of $\ell_1(\kappa^+)$.

2 Definitions and Source Input

The source paper defines a normed space X to be κ -octahedral if for every subspace $Y \subset X$ with $\text{dens}(Y) \leq \kappa$ and every $\varepsilon > 0$, there is $x \in S_X$ such that

$$\|\lambda x + y\| \geq (1 - \varepsilon)(|\lambda| + \|y\|) \quad (\lambda \in \mathbb{R}, y \in Y).$$

We use the construction from Theorem 5.13 of [1]. Fix an infinite cardinal κ , put $\Gamma = \kappa^+$, choose $\alpha_n \in (1/2, 1)$ with $\sup_n \alpha_n = 1$, and set

$$p_n = (1 + \log_2(\alpha_n))^{-1} \in (1, \infty), \quad X_n = \ell_{p_n}(\Gamma), \quad X = \left(\bigoplus_{n=1}^{\infty} X_n \right)_{\ell_1}.$$

Theorem 5.13 of [1] proves that this space X is κ -octahedral.

3 Proof Intuition

The source construction is an ℓ_1 -sum of reflexive blocks whose exponents approach the endpoint needed for κ -octahedrality. The possible embedding of $\ell_1(\kappa^+)$ has one vector Te_γ for each $\gamma < \kappa^+$. Since each individual vector in an ℓ_1 -sum has small tail after sufficiently many blocks, and since κ^+ is a regular uncountable cardinal, a subfamily of size κ^+ has a common finite tail cutoff. On that subfamily the finite initial block projection remains bounded below. But a finite sum of the blocks is reflexive, so it cannot contain ℓ_1 of any infinite index set. This rules out the embedding.

4 Counterexample

Theorem 1. *For every infinite cardinal κ there exists a κ -octahedral Banach space which contains no isomorphic copy of $\ell_1(\kappa^+)$. Consequently, Question 6.2 of [1] has a negative answer, already for $\kappa = \omega$.*

Proof. Let X be the space displayed above:

$$X = \left(\bigoplus_{n=1}^{\infty} \ell_{p_n}(\kappa^+) \right)_{\ell_1}, \quad 1 < p_n < \infty.$$

By Theorem 5.13 of [1], X is κ -octahedral. It remains to show that X does not contain an isomorphic copy of $\ell_1(\kappa^+)$.

Write $\Gamma = \kappa^+$ and suppose, toward a contradiction, that $T : \ell_1(\Gamma) \rightarrow X$ is an isomorphic embedding. Choose $c > 0$ such that

$$\|Tu\| \geq c\|u\|_{\ell_1(\Gamma)} \quad (u \in \ell_1(\Gamma)).$$

For $N \in \mathbb{N}$, let $P_N : X \rightarrow \bigoplus_{n=1}^N \ell_{p_n}(\Gamma)$ be the coordinate projection onto the first N summands and set $R_N = I - P_N$. Fix η with $0 < \eta < c$. For every $\gamma \in \Gamma$, the vector Te_γ belongs to the countable ℓ_1 -sum X , hence there is $N_\gamma \in \mathbb{N}$ such that

$$\|R_{N_\gamma}Te_\gamma\| < \eta.$$

Since the successor cardinal $\Gamma = \kappa^+$ is regular and uncountable, one of the sets

$$\Lambda_N = \{\gamma \in \Gamma : N_\gamma \leq N\}$$

has cardinality Γ . Fix such an N and put $\Lambda = \Lambda_N$.

For every finitely supported $u = \sum_{\gamma \in F} a_\gamma e_\gamma$ with $F \subset \Lambda$, we have

$$\|R_N Tu\| \leq \sum_{\gamma \in F} |a_\gamma| \|R_N Te_\gamma\| \leq \sum_{\gamma \in F} |a_\gamma| \|R_{N_\gamma} Te_\gamma\| < \eta \sum_{\gamma \in F} |a_\gamma| = \eta \|u\|_{\ell_1}.$$

Therefore

$$\|P_N Tu\| \geq \|Tu\| - \|R_N Tu\| \geq (c - \eta)\|u\|_{\ell_1}.$$

By density of finitely supported vectors, $P_N T$ restricts to an isomorphic embedding of $\ell_1(\Lambda)$ into the finite direct sum

$$F_N = \left(\bigoplus_{n=1}^N \ell_{p_n}(\Gamma) \right)_{\ell_1}.$$

Each space $\ell_{p_n}(\Gamma)$ is reflexive because $1 < p_n < \infty$, and a finite direct sum of reflexive spaces is reflexive. Thus F_N is reflexive. Its closed subspace $(P_N T)(\ell_1(\Lambda))$ would also be reflexive, which is impossible because $\ell_1(\Lambda)$ is nonreflexive for every infinite Λ . This contradiction proves that no such T exists. $\square$

5 Verification Notes

The proof is non-computational. The only code in the packet is `code/make_crop.py`, which regenerates the screenshot from `source_paper.pdf`. The source PDF was compiled from the locally ingested arXiv source for 2102.13525. The finite-block argument uses three standard facts: successor cardinals are regular, $\ell_p(\Gamma)$ is reflexive for $1 < p < \infty$, and closed subspaces of reflexive Banach spaces are reflexive.

6 Limitations and Novelty Check

This packet answers only Question 6.2 of [1]. It does not answer Question 6.1 about equivalent renormings failing the $(-1)\text{-BCP}_\kappa$. The counterexample uses exactly the normed space constructed in Theorem 5.13 of the source paper, with the additional observation that it cannot contain $\ell_1(\kappa^+)$.

Before promotion, the lightweight run indexes were searched for the arXiv id 2102.13525, for `kappa-octahedral`, and for close `ell_1(kappa+)` phrases. External exact-phrase web searches on 2026-07-01 for the source title, Question 6.2, `kappa-octahedral`, and `ell_1(kappa+)` found only the source paper or irrelevant hits, not a later exact answer. Novelty confidence is therefore bounded but positive.

7 Human Review Recommendation

Review the proof of noncontainment, especially the use of regularity of κ^+ to obtain a common finite cutoff for κ^+ many basis vectors. If that step is accepted, the source theorem supplies κ -octahedrality and the packet gives a direct counterexample to Question 6.2.

References

- [1] S. Ciaci, J. Langemets, and A. Lissitsin, *A Characterization of Banach Spaces Containing $\ell_1(\kappa)$ via ball-covering properties*, arXiv:2102.13525, 2021.